



Tiling Window Managers

A window manager is perhaps the most ubiquitous yet invisible aspect of any modern desktop operating system. This software manages the placement of windows and dictates where newly opened windows will show up. Whenever you resize, move, minimise/maximise a window, or click to focus it, you are interacting with a window manager. This article discusses floating and tiling window managers.

Here is a list of a few popular window managers used on Linux desktops:

Desktop environment	Window manager
KDE	Kwin
GNOME 2	Metacity
GNOME 3	Mutter
Unity	Compiz
XFCE	Xfwm
LXDE	Openbox

It is worth mentioning that a window manager (WM) is only concerned with managing existing windows and handling newly opened ones. How these windows are opened is of no concern to the WM. Hence, desktop environments like KDE, GNOME, etc, often provide extra applications like panels, application launchers, system trays, file managers, etc, to provide a complete and consistent set of tools to the end user.

It shouldn't come as a surprise that there are a variety of WMs. In this article, I will cover two kinds of WMs—floating and (surprise!) tiling. All the WMs mentioned in the table are the floating variety. As the name implies, they represent windows as floating rectangles on the desktop—freely resizable and movable. The floating approach to window management has been the most widely used (even Windows and Mac use it), but it has some fundamental flaws, which most of

you might have experienced at some point or the other.

Problems with conventional floating WMs

The main problem with floating WMs is the fact that they are not really designed to facilitate side-by-side viewing of more than one window. The idea of having freely floating windows means that they can (and will) overlap each other. Having to manually move and resize windows to prevent them from covering the mission-critical parts of one another can get pretty tiresome, once you have more than a handful open. Even if you have all the open windows arranged the way you want, you will still have to go through the resize-move-rinse-repeat rigmarole each time you add or remove windows from the layout.

Another problem is the over-reliance on the mouse. Anyone with a display larger than, say, 45.72 cm (18 inches) would have felt the pain of having to move the mouse from one corner of the screen to another, and then back to the initial position—a fairly typical motion on a Windows PC, where the *Start* menu is in the bottom-left corner of the screen, and the *min/max/close* buttons at the top-right corner of a maximised window. Generally, the larger the monitor, the more you will find the need to get acquainted with keyboard short-cuts instead of using the mouse. These are precisely the areas where tiling WMs shine.

An introduction to tiling WMs

A tiling WM does not place windows as floating entities, but as non-overlapping rectangular tiles (think of tiles on a